



Meditation & Spiritual Audio Streaming App

Full Project Documentation Report

Anurag Rishi — Find Peace Within

Report Date: 21 June 2026

Version 1.0 — Release Build

1. Project Overview

Know Thyself (formerly Anurag Rishi) is a Flutter-based mobile application for Android that delivers meditation, spiritual talks, and guided audio sessions. The app features a fully dark-violet themed UI, offline download capabilities with AES-256-CTR encryption, a one-device-per-account lock for DRM, background audio playback, Supabase-powered backend, and an admin dashboard for user/content management.

App Name	Know Thyself
Previous Name	Anurag Rishi
Platform	Android (Flutter)
Backend	Supabase (PostgreSQL + Edge Functions + Auth + Storage)
Storage	Cloudflare R2 (private bucket, signed URLs)
Build Target	Release APK — flutter build apk --release
Design Theme	Dark violet (#12082E background, #8B5CF6 accent, #EC4899 pink)
Architecture	Feature-first, Riverpod state management, Repository pattern

2. Technology Stack

2.1 Frontend — Flutter

Flutter	3.x — cross-platform UI toolkit
Dart	3.3+ — language with null safety
flutter_riverpod	^2.5.1 — reactive state management (Providers, FutureProviders, StateProviders)
go_router	^14.2.0 — declarative routing with auth redirect guards
just_audio	^0.9.40 — audio playback engine
audio_service	^0.18.15 — background audio + Android notification controls
flutter_secure_storage	^9.2.2 — encrypted key-value store for device salt & download keys
device_info_plus	^10.1.0 — hardware device identifiers for fingerprinting
path_provider	^2.1.4 — filesystem paths for encrypted download storage
http	^1.2.2 — HTTP client for proxy requests
pointycastle	^3.9.1 — AES-256-CTR decryption for offline playback
crypto	^3.0.3 — SHA-256 device fingerprint hashing
rxdart	^0.27.7 — stream composition for audio state
flutter_launcher_icons	^0.13.1 — automated launcher icon generation

2.2 Backend — Supabase

Supabase Auth	Email/password authentication, session management, JWT tokens
PostgreSQL	Primary database — users, audio content, categories, subscriptions, devices
Row Level Security	RLS policies ensuring users only access their own data
Edge Functions	Deno serverless functions for audio stream signing, access verification
Realtime	Auth state change streams consumed by GoRouter redirect guard
Project Ref	gzcanqovqirarnculjqj

2.3 Storage — Cloudflare R2

Bucket Type	Private — no public access
Access Method	Time-limited signed URLs generated by Supabase Edge Functions
Content	Encrypted audio files (MP3/M4A)
Security	Audio URLs expire after short window; re-signed on each play request

2.4 Architecture Pattern

- Feature-first folder structure: each feature has its own domain / data / presentation / application layers
- Repository pattern: abstract interfaces in domain, concrete implementations in data layer
- Riverpod Providers wire everything together — no service locators or singletons
- GoRouter handles all navigation with declarative route guards based on auth state

3. Features Built

3.1 Splash Screen

- Dark violet radial gradient background
- Glowing lotus logo with animated fade-in and scale-up (1.4 s animation)
- "Anurag Rishi / Find Peace Within" wordmark centred on screen
- Auto-routes to /home or /login after 2.6 s based on active session
- Router guard prevents mid-animation redirect away from splash

3.2 Authentication

- Login screen with email/password via Supabase Auth
- Forgot Password screen
- Auth state streamed into GoRouter — automatic redirect on login/logout
- One-device-per-account lock: device registers on login; second device blocked
- Admin can "Reset Device" to allow user to re-register on new device

3.3 Home Screen

- Dark violet themed — background #12082E, cards #221550
- Real username greeting pulled from Supabase profile
- Daily Card: first featured audio with cover art and Play Now button
- Featured For You: horizontal scrollable row of audio cards
- Categories row: colour-coded icons, tap navigates to filtered browse
- Continue Listening bar: resumes last in-progress audio with progress indicator
- Search bar navigates to Browse/Search screen

3.4 Browse / Search Screen

- Dual mode: search (with live text field) or category filter
- Audio rows with gradient thumbnail, title, artist, duration
- Tap any audio — starts streaming immediately, navigates to Now Playing

3.5 Now Playing Screen

- Full dark violet design with glowing album art ring
- Seek bar with mm:ss / h:mm:ss formatting
- Play / Pause / Previous / Next transport controls
- Repeat and Shuffle toggles (highlight violet when active)
- Favourite heart toggle (pink when active)
- Sleep timer with countdown display

- Playback speed selector (0.5× – 2×)
- Download button to save audio for offline use

3.6 Offline Downloads

- AES-256-CTR encryption — audio stored encrypted on device, never as plaintext
- Loopback decrypting proxy (127.0.0.1) serves decrypted bytes to just_audio
- Downloads screen lists all downloads with live status (queued / downloading / paused / completed / failed)
- Pause, resume, delete actions per download
- Offline Player screen: plays encrypted local file, seek bar, play/pause — fully dark themed
- Downloads survive app restart (metadata persisted, restored on boot)
- Expired/revoked downloads purged automatically on startup

3.7 Profile Screen

- Dark violet circular gradient avatar
- Display name + plan badge (■ Premium Member)
- Stats row: Day Streak / Minutes / Sessions
- Achievements section with 4 badge icons (■ ■ ■ ■)
- Subscription row showing plan name
- Settings button opens bottom sheet with: Logout, Device Info

3.8 Mini Player (Footer Bar)

- Persistent across Home, Downloads, Profile screens via AppShell wrapper
- Dark violet surface (#1C1040) — matches app theme
- Shows album art thumbnail, title, artist, play/pause button
- Thin violet progress bar at top
- Tap navigates to full Now Playing screen

3.9 App Identity

- App name: Know Thyself (AndroidManifest + AppConfig)
- Launcher icon: custom lotus image (assets/app_icon.png via flutter_launcher_icons)
- Adaptive icon with dark violet (#12082E) background

4. Database Schema (Supabase / PostgreSQL)

profiles	User profile — displayName, email, role, subscriptionTier, country, status, avatarUrl, access_exp
audios	Audio content — title, artist, cover_art_url, duration_seconds, is_premium, status, play_count, cr
categories	Browse categories — name, slug, sort_order
audio_categories	Many-to-many join: audios ↔ categories
subscriptions	User subscriptions — plan, status, current_period_end, cancel_at_period_end
subscription_plans	Plan definitions — name, price, duration
devices	Registered devices — user_id, device_fingerprint, device_name, platform, is_active
watch_history	Playback progress — user_id, audio_id, progress_seconds, duration_seconds, completed, last_
favorites	User favourite audio items

Key RPCs / Edge Functions

- has_active_access() — server-side RPC checks if user has valid access window
- register_device — validates one-device lock, registers new device fingerprint
- Audio stream signing edge function — generates time-limited Cloudflare R2 signed URL

5. Problems Faced & Solutions

5.1 Cross-Drive Kotlin Build Crash (Windows)

Problem	Project on D: drive, Flutter pub cache on C: drive. Kotlin incremental compiler failed with "this and base files ha
Solution	Added kotlin.incremental=false to mobile_app/android/gradle.properties. This disables Kotlin's incremental con

5.2 Audio Not Playing After Device Reset

Problem	After admin used "Reset All Devices" in the admin panel, all users lost audio/download access. _getActiveDevi
Solution	The fix is operational: users must log out and log back in after a device reset. Re-login triggers register_device

5.3 Specific User (test@test.com) Always Failing

Problem	One test account had a stale active device row from a different emulator/physical device. Even after reset, the
Solution	Admin panel → ... menu → "Reset device" on the specific user → user logs back in. Added per-user reset butt

5.4 Splash Screen Not Centred

Problem	Opening screen used Spacer(flex:5) above/below lotus and another Spacer(flex:2) above the wordmark, pushi
Solution	Replaced the Spacer-based Column with a single Center widget wrapping all elements in a Column(mainAxisS

5.5 Mini Player Footer White / Wrong Theme

Problem	MiniPlayer widget used AppTheme.surfaceElevated (light cream colour) as its background, making the footer b
Solution	Replaced AppTheme references in mini_player.dart with hardcoded dark violet constants (#1C1040 backgroun

5.6 num vs double Type Error in Seek Bar

Problem	Dart's double.clamp(double, double) returns num, not double. The Slider widget requires a double for its value.
Solution	Added .toDouble() after the clamp() call, then cast the result as double. This satisfies Dart's strict type system v

5.7 Git Merge Conflicts on gradle.properties

Problem	Local Windows machine had uncommitted edits to gradle.properties (the kotlin.incremental=false fix). git pull fa
Solution	Used git stash → pull → stash pop workflow. When vim opened for merge message, used :wq to save and exit

5.8 UI Regression After Revert

Problem	Early in the project, UI redesign caused lag/instability. Full revert to commit 991734c was performed, which also
Solution	Recreated browse_screen.dart from scratch with the dark violet theme. Router was updated to add /search and

6. Security Model

6.1 One-Device Lock (DRM)

- On login, app generates a SHA-256 device fingerprint from hardware ID + secure random salt
- Salt stored in FlutterSecureStorage (survives reinstall on iOS Keychain)
- register_device RPC checks if another active device exists for the user
- If yes, registration is denied — user sees "already logged in on another device"
- Admin can reset a user's device from the admin panel

6.2 Encrypted Offline Storage

- Audio files downloaded and stored with AES-256-CTR encryption
- Encryption key unique per download, stored in FlutterSecureStorage
- Playback: local loopback proxy (127.0.0.1) decrypts on-the-fly into just_audio
- No plaintext audio file ever written to disk
- On subscription expiry, purgeRevokedAndExpired() deletes local files

6.3 Network Security

- Cloudflare R2 bucket is private — no public URLs
- Time-limited signed URLs generated per request by Supabase Edge Function
- Supabase RLS policies enforce per-user data isolation at database level
- has_active_access() RPC validated server-side before any stream URL is issued

7. Project File Structure

```
mobile_app/ ■■■ android/ ■ ■■■ app/src/main/AndroidManifest.xml ← app name,
permissions ■ ■■■ gradle.properties ← kotlin.incremental=false fix ■■■ assets/ ■
■■■ app_icon.png ← launcher icon (lotus) ■■■ lib/ ■ ■■■ main.dart ← app entry,
Riverpod scope, theme ■ ■■■ app/ ■ ■ ■■■ router/app_router.dart ← GoRouter, auth
guards, all routes ■ ■ ■■■ theme/app_theme.dart ← light + dark ThemeData ■ ■
■■■ widgets/ ■ ■ ■■■ app_shell.dart ← wraps screens with MiniPlayer ■ ■ ■■■
lotus_logo.dart ← custom lotus SVG widget ■ ■■■ core/ ■ ■ ■■■
config/app_config.dart ← Supabase URL, keys, app name ■ ■ ■■■
device/device_info_service.dart ■ ■ ■■■ network/supabase_client_provider.dart ■
■■■ features/ ■ ■■■ auth/ ← login, splash, auth providers ■ ■■■ audio/ ← player
handler, now playing, mini player ■ ■■■ downloads/ ← encrypted downloads, offline
player ■ ■■■ home/ ← home screen, browse, search ■ ■■■ profile/ ← profile
screen, settings sheet supabase/ ■■■ functions/ ← Deno edge functions (stream
signing) PROJECT_LOG.md ← full operational log
```

8. Screens Summary

Screen	Route	Key Components
Splash	/splash	Animated lotus, auto-navigate to home/login
Login	/login	Email/password, forgot password link
Forgot Password	/forgot-password	Supabase password reset email
Home	/home	Greeting, daily card, featured, categories, continue listening
Browse / Search	/search	Live text search over audios
Category Browse	/category/:id	Filtered audio list by category
Now Playing	/now-playing	Full player, seek, controls, download, timer
Downloads	/downloads	Download queue with status and actions
Offline Player	/offline-player/:id	Encrypted local playback
Profile	/profile	Avatar, stats, achievements, settings sheet

9. Repository & Version Control

Repository	github.com/cyberking-coder/Rishi_app2
Working Branch	claude/funny-keller-sbua4r
Base Commit	991734c — Last stable state (pre-redesign)
Final Commit	b65c129+ — Know Thyself full release
Key Commits	991734c Revert to stable → splash → now playing → home → profile → bug fixes → rename

All development was done on the feature branch `claude/funny-keller-sbua4r`. The branch was built incrementally: revert to stable baseline → splash screen → now playing redesign → home screen redesign → browse/search → profile redesign → theme fixes (downloads, mini player, offline player) → app rename → launcher icon setup.

10. Build & Run Instructions

Prerequisites

- Flutter SDK 3.x installed and on PATH
- Android SDK with a connected device or emulator
- Java 17+ (for Gradle)
- Git

Clone & Setup

```
git clone https://github.com/cyberking-coder/Rishi_app2.git cd Rishi_app2 git checkout claudes/funny-keller-sbua4r cd mobile_app flutter pub get
```

Generate Launcher Icons (one-time)

```
dart run flutter_launcher_icons
```

Debug Run

```
flutter run
```

Release APK

```
flutter build apk --release # Output:  
build/app/outputs/flutter-apk/app-release.apk
```

Windows-specific Note

If the project and Flutter pub cache are on different drives (e.g. project on D:, pub cache on C:), Kotlin incremental compilation will crash. The fix is already committed — `gradle.properties` contains `kotlin.incremental=false`.

11. Known Limitations & Future Work

Day Streak tracking	Always shows 0 — daily usage not tracked in watch_history. Needs a daily_streaks table or RPC.
Minutes / Sessions stats	Always show 0 — aggregate queries not yet implemented. Can be derived from watch_history.
Gender-based avatar	Profile uses a generic person icon. UserProfile entity has no gender field. Would need a DB column.
iOS support	Not configured — no iOS provisioning profile, no Info.plist changes. Straightforward to add.
Gradle version warnings	AGP 8.9.1 and Gradle 8.12.0 are below Flutter's upcoming minimum. Should upgrade to AGP 8.10.0.
Push notifications	Not implemented. Could use Supabase Realtime or Firebase Cloud Messaging for new content.
Chromecast / AirPlay	Not implemented. just_audio supports cast on some platforms with additional setup.

Know Thyself

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